

All measure metric for Regression Model

Key Terms and Formulas

- y_i = actual (observed) value for the i-th observation
- $\hat{y}_i$ = predicted value from the model for the i-th observation
- $\bar{y}$ = mean of all actual y values
- n = number of observations
- p = number of predictors (independent variables) + 1 (for the intercept)

METRIC	FULL NAME	FORMULA.	WHAT IT MEASURES	UNITS	INTERPRETATION / USE CASE
RSS	Residual Sum of Squares	$RSS = \sum (y_i - \hat{y}_i)^2$	Total squared error between actual and predicted values	Same as y^2	Raw measure of total error. Lower = better fit. Basis for many other metrics.
TSS	Total Sum of Squares	$TSS = \sum (y_i - \bar{y})^2$	Total variability in the dependent variable (around its mean)	Same as y^2	Represents total variation in data. Used as denominator in R^2 .
ESS / SSR	Explained / Regression Sum of Squares	$SSR = \sum (\hat{y}_i - \bar{y})^2$	Amount of variability explained by the model	Same as y^2	Higher = model explains more variability. $TSS = SSR + RSS$ (always true).
MSE	Mean Squared Error	$MSE = RSS / n$ or sometimes $MSE = RSS / (n - p)$ (unbiased version)	Average squared error per observation	Same as y^2	Popular in machine learning. Lower = better. Penalizes large errors heavily.
RSE / RMSE	Residual Standard Error / Root Mean Squared Error	$RSE = \sqrt{RSS / (n - p)}$ $RMSE = \sqrt{MSE} = \sqrt{RSS / n}$	Square root of MSE — "typical" size of prediction error	Same as y	Most interpretable error metric (same units as y). Lower = better predictions.

METRIC	FULL NAME	FORMULA.	WHAT IT MEASURES	UNITS	INTERPRETATION / USE CASE
MAE	Mean Absolute Error	$MAE = (1/n) \sum$	$y_i - \hat{y}_i$		Average absolute error